Sec: SR.IIT_*COSC(MODEL-A&APEX) | RTA-2 | Date: 15-02-26
Time: 3 Hrs | 2023_P1 | Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**ACD**	2	**ABD**	3	**BC**	4	**B**	5	**C**
6	**C**	7	**C**	8	**3**	9	**74**	10	**5**
11	**1**	12	**25**	13	**24**	14	**C**	15	**C**
16	**C**	17	**B**						

PHYSICS

18	**ABC**	19	**AC**	20	**AC**	21	**B**	22	**A**
23	**B**	24	**B**	25	**2**	26	**37**	27	**250 TO 255**
28	**11**	29	**2**	30	**6**	31	**C**	32	**D**
33	**D**	34	**D**						

CHEMISTRY

35	**ABD**	36	**C**	37	**ACD**	38	**C**	39	**D**
40	**C**	41	**A**	42	**6**	43	**5**	44	**4**
45	**2**	46	**16**	47	**6**	48	**A**	49	**B**
50	**A**	51	**A**						

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SOLUTIONS

MATHS

1. $\because \dfrac{x^4+9}{x^6(x^2-9)} = \dfrac{1}{x^2(x^2-9)} + \dfrac{9}{729}\left(\dfrac{x^6-(x^6-243)}{x^6(x^2-9)}\right)$

$= \dfrac{5}{243}\left(\dfrac{1}{x-3} - \dfrac{1}{x+3}\right) - \dfrac{10}{81x^2} - \dfrac{1}{9x^4} - \dfrac{1}{x^6}$

Now integrate

$\Rightarrow \dfrac{5}{243}\ell n\left|\dfrac{x-3}{x+3}\right| + \dfrac{10}{81x} + \dfrac{1}{27x^3} + \dfrac{1}{5x^5} + k$

2. Area of region $OBA = \int_0^h f(x)dx - \int_0^h \sqrt{x}\,dx$

and area of region OCA = area of rectangle

$-\int_0^h f(x)dx = f(h).h - \int_0^h f(x)dx$

$\Rightarrow \int_0^h f(x)dx - \int_0^h \sqrt{x}dx = f(h).h - \int_0^h f(x)dx$

$\Rightarrow 2\int_0^h f(x)dx = \dfrac{2}{3}x^{\frac{3}{2}}\Big|_0^h + hf(h)$

differentiate both side w.r.t. h

$\Rightarrow 2f(h) = h^{\frac{1}{2}} + hf'(h) + f(h)$

$\Rightarrow 2y = \sqrt{x} + x\dfrac{dy}{dx} + y\left(\because f(h) = y, h = x\right)$

$\Rightarrow \dfrac{dy}{dx} - \left(\dfrac{1}{x}\right)y = \dfrac{-1}{\sqrt{x}},$

3. $f(x+y) = f(x) + f(y) + \dfrac{x+y}{xy} - \dfrac{1}{x+y}$ …(1)

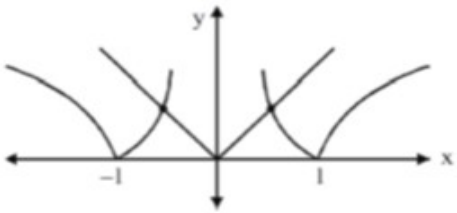
Partially differentiating w.r.t. to x

$f'(x+y) = f'(x) - \dfrac{1}{x^2} + \dfrac{1}{(x+y)^2}$

Putting x = 1

$f'(1+y) = 2 - 1 + \dfrac{1}{(1+y)^2}$

$\Rightarrow f'(1+y) = 1 + \dfrac{1}{(1+y)^2} \Rightarrow f(x) = 1 + \dfrac{1}{x^2}$



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$\Rightarrow f(x) = x - \frac{1}{x} + C$

Put $x = y = 1, f(2) - 2f(1) = 2 - \frac{1}{2} = \frac{3}{2}$

Hence $f(x) = x - \frac{1}{x}, C = 0$

i) $\int_1^e f(x)dx = \int_1^e \left(x - \frac{1}{x}\right)dx = \left(\frac{x^2}{2} - \ell nx\right)_1^e \Rightarrow \frac{e^2 - 3}{2}$

(ii) $\left|f(|x|)\right| = |x|$

number of solution = 2

(iii) $g(x) = \frac{f(x)}{x^5 - \frac{1}{x}} = \frac{x - \frac{1}{x}}{x^5 - \frac{1}{x}} = \frac{x^2 - 1}{x^6 - 1}$

$g(x) = \frac{1}{x^4 + x^2 + 1} = (x \neq 0, \pm 1)$

Range of g(x) is (0, 1) - $\left\{\frac{1}{3}\right\}$

4. $\frac{dy}{dx} + y^3(1 + \cos x) - \frac{y}{x} = 0$

$\frac{1}{y^3}\frac{dy}{dx} - \frac{1}{y^2 x} = -(1 + \cos x)$

Let $\frac{-1}{y^2} = u$

$\therefore \frac{1}{2}\frac{du}{dx} + \frac{u}{x} = -(1 + \cos)$

IF = $e^{\int \frac{2}{x}dx = x^2}$

$\therefore ux^2 = -2\int x^2(1 + \cos x)dx$

$\Rightarrow \frac{x^2}{2y^2} = \frac{x^3}{3} + x^2 \sin x + 2x\cos x - 2\sin x + k$

5. $y = \{x\}^{[x]} + [x]$

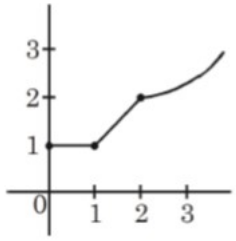
$$y = \begin{cases} 1 & x \in [0,1) \\ x & x \in [1,2) \\ (x-2)^2 + 2 & x \in [2,3) \\ 3 & x = 3 \end{cases}$$

$A = 1 + \frac{3}{2} + \int_2^3 \left((x-2)^2 + 2\right)dx$

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$$A=\frac{5}{2}+\left(\frac{(x-2)^3}{3}+2x\right)_2^3 \Rightarrow A=\frac{5}{2}+\left(\frac{1}{3}+6\right)-4$$

$$=\frac{29}{6}$$



6. $$\sum_{k=1}^{\infty}\left(\frac{1}{5k-1}-\frac{1}{5k}\right)=\sum_{k=1}^{\infty}\int_0^1\left(x^{5k-2}-x^{5k-1}\right)dx$$

$$\int_0^1\sum_{k=1}^{\infty}x^{5k-2}(1-x).dx$$

$$\int_0^1\frac{x^3}{1+x+x^2+x^3+x^4}$$

7. We have $\underbrace{\left|\left(|x|-2\right)^2+1\right|}_{\geq 1}+\underbrace{\left|x^3-4x\right|+3}_{\geq 3}=4$

Minimum value of L.H.S. is 4 when

$$\left.\begin{array}{rcl}|x|-2=0 & \Rightarrow & x=\pm 2\\ x^3-4x=0 & \Rightarrow & x=0,\pm 2\end{array}\right\}\Rightarrow x=2 \text{ or } x=-2$$

Since $x_1<x_2\Rightarrow x_1=-2 \text{ or } x_2=2$ which satisfy the given equation

Also, $\int_{\frac{1}{2}}^{\frac{3}{2}}f(x)dx=6\Rightarrow\int_0^1 f(x)dx=6$

Now, $\int_{-2}^{2}f(x)dx=4\int_0^1 f(x)dx=24$

8. $$\frac{dy}{dx}=\frac{4y^3+2yx^2}{3xy^2+x^3},y(1)=1$$

$$\frac{dy}{dx}=\frac{4(y/x)^3+2(y/x)}{3(y/x)^2+1}$$

$$y=xp$$

$$x\frac{dp}{dx}+p=\frac{4p^3+2p}{3p^2+1}\Rightarrow x\frac{dp}{dx}=\frac{p^3+p}{3p^2+1}$$

$$\int\frac{3p^2+1}{p^3+p}dp=\int\frac{dx}{x}$$

$$\ell n\left(p^3+p\right)=\ell nx+\ell nC$$

$$p^3+p=xC$$

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$$\left(\frac{y}{x}\right)^3+\left(\frac{y}{x}\right)=xC$$

$$y^3+x^2y=x^4C$$

$$x=1,y=1$$

$$1+1=C\Rightarrow C=2$$

$$y^3+x^2y=2x^4$$

Put x = 2

$$y^3+4y-32=0$$

Having root between 2 and 3

$$y(2)\in[2,3)$$

9. $$\frac{1}{e-1}\sum_{n=1}^{73}\frac{1}{(n+1)!}\int_0^n e^{\{x\}}dx$$ (Using King)

$$=\frac{1}{e-1}\sum_{n=1}^{73}\frac{n(e-1)}{(n+1)!}=\sum_{n=1}^{73}\frac{1}{n!}-\frac{1}{(n+1)!}$$

$$=1-\frac{1}{74!}\Rightarrow m=74!$$

Number of Cyphers $=\left[\frac{74}{5}\right]+\left[\frac{74}{5^2}\right]=14+2=16$

10. $$\int\frac{1}{x^{5/6}(x-2)^{7/2}}dx-\int\frac{1}{x^{3/4}(x-2)^{5/4}}dx$$

$$=3\sqrt[6]{\frac{x}{x-2}}-2.\sqrt[4]{\frac{x}{x-2}}+c;$$

$$\therefore f(0)=0\Rightarrow c=0$$

$$f(3)=3.\sqrt[6]{3}-2.\sqrt[4]{3}\qquad\Rightarrow A=3$$

$$B=-2\qquad |AB|=6$$

11. $$I=\int_0^{\frac{\pi}{2}}\frac{\cos^4x+\sin x\cos^3x+\sin^2x\cos^2x+\sin^3x\cos x}{\sin^4x+\cos^4x+2\sin x\cos^3x+2\sin^2x\cos^2x+2\sin^3x\cos x}dx$$

$$I=\int_0^{\frac{\pi}{2}}\frac{(\sin x\cos^3x+\sin^3x\cos x)+(\cos^4x+\sin^2x\cos^2x)}{\sin^4x+\cos^4x+2\sin^2x\cos^2x+(2\sin x\cos^3x+2\sin^3x\cos x)}dx$$

$$=\int_0^{\frac{\pi}{2}}\frac{\sin x\cos x+(\cos^4x+\sin^2x\cos^2x)}{\sin^4x+\cos^4x+2\sin^2x\cos^2x+2\sin x\cos x}dx$$

Use King and add

$$2I=\int_0^{\frac{\pi}{2}}\frac{\sin^4x\cos^4x+2\sin^2x\cos^2x+2\sin x\cos x}{\sin^4x+\cos^4x+2\sin^2x\cos^2x+2\sin x\cos x}dx=\frac{\pi}{2}$$

$$\therefore I=\frac{\pi}{4}$$

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12. $$\frac{x}{4x^2-2x+9}=\frac{1}{4x+\frac{9}{x}-2}\le\frac{1}{10}$$

$$\left\{\because 4x+\frac{9}{x}\ge 12 \text{ here } x\in\left[\frac{1}{5},5\right]\right\}$$

$$I=\underbrace{\int_{1/5}^{5}\frac{1}{x}\left\{\frac{x}{4x^2-2x+9}\right\}dx}_{I_1}+\underbrace{\int_{1/5}^{5}\frac{1}{x}\left\{\frac{x}{9x^2-2x+4}\right\}dx}_{I_2}$$

Putting $x=\frac{1}{t}\Rightarrow dx=-\frac{1}{t^2}dt$

$$I_2=-\int_{5}^{1/5}t\left\{\frac{\frac{1}{t}}{\frac{9}{t^2}-\frac{2}{t}+4}\right\}\frac{1}{t^2}dt=\int_{1/5}^{5}\frac{1}{t}\left\{\frac{t}{4t^2-2t+9}\right\}dt$$

Now, $I=\int_{1/5}^{5}\frac{1}{x}dx \quad \{\because \{x\}+\{-x\}=1 \text{ when } x\notin I\}$

13. $$(P)\left(1+\left(\frac{dy}{dx}\right)^2\right)^3=\lambda\left(\frac{d^2y}{dx^2}\right)^2,\ \alpha=2,\ \beta=2$$

$$(\alpha+\beta)=4$$

(Q) $$y=\tan\left(\frac{\pi}{4}+ax\right)\tan\left(\frac{\pi}{4}-ax\right)+ce^{bx+d}$$

$$y=\left(\frac{1+\tan ax}{1-\tan ax}\right)\left(\frac{1-\tan ax}{1+\tan ax}\right)+ce^{d}.e^{bx}$$

$$y=1+A.B^{x}, \qquad \text{order}=2$$

(R) $$\frac{dy}{dx}=y-y^2\Rightarrow\frac{dy}{y(1-y)}=dx$$

$$\log y-\log|1-y|=x+C$$

$$\log 2-\log 1=0+C$$

$$C=\log_e 2$$

$$\log y-\log|y-1|=x+\log_e 2$$

$$\log\left|\frac{y}{y-1}\right|=\log_e 2+x$$

$$\frac{y}{y-1}=2e^{x}\Rightarrow y=2e^{x}y-2-2e^{x}$$

$$y=\frac{2e^{x}}{2e^{x}-1}\Rightarrow\lim_{x\to\infty}y=1$$

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(S) $x = my + \frac{1}{m}$

$\frac{dx}{dy} = m \Rightarrow m = \frac{1}{y'}$

$x = \frac{y}{y'} + y' \Rightarrow (y')^2 + y = xy'$

$\alpha = 1,\ \beta = 2$

$\alpha + \beta = 3$

14. $(P)\left(1+\left(\frac{dy}{dx}\right)^2\right)^3 = \lambda\left(\frac{d^2y}{dx^2}\right)^2,\ \alpha = 2,\ \beta = 2$

$(\alpha + \beta) = 4$

(Q) $y = \tan\left(\frac{\pi}{4} + ax\right)\tan\left(\frac{\pi}{4} - ax\right) + ce^{bx+d}$

$y = \left(\frac{1+\tan ax}{1-\tan ax}\right)\left(\frac{1-\tan ax}{1+\tan ax}\right) + ce^d.e^{bx}$

$y = 1 + A.B^x,$ order = 2

(R) $\frac{dy}{dx} = y - y^2 \Rightarrow \frac{dy}{y(1-y)} = dx$

$\log y - \log|1-y| = x + C$

$\log 2 - \log 1 = 0 + C$

$C = \log_e 2$

$\log y - \log|y-1| = x + \log_e 2$

$\log\left|\frac{y}{y-1}\right| = \log_e 2 + x$

$\frac{y}{y-1} = 2e^x \Rightarrow y = 2e^x y - 2 - 2e^x$

$y = \frac{2e^x}{2e^x - 1} \Rightarrow \lim_{x\to\infty} y = 1$

(S) $x = my + \frac{1}{m}$

$\frac{dx}{dy} = m \Rightarrow m = \frac{1}{y'}$

$x = \frac{y}{y'} + y' \Rightarrow (y')^2 + y = xy'$

$\alpha = 1,\ \beta = 2$

$\alpha + \beta = 3$

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15. $(P)\ 2x^4ydy + y^2dy = x^2dx - 4x^3y^2dx$

$x^4(2y)dy + y^2(4x^3dx) + y^2dy - x^2dx = 0$

$d(x^4.y^2) + \frac{1}{3}d(y^3) - \frac{1}{3}d(x^3) = 0$

$x^4y^2 + \frac{y^3}{3} - \frac{x^3}{3} = c$

$3x^4y^2 + y^3 - x^3 = 3c$

(Q) Apply componendo & dividend

$\frac{e^x}{e^{-x}} = \frac{dx}{dy} \Rightarrow dy = e^{-2x}dx \Rightarrow y = \frac{-e^{-2x}}{2} + c$

$2ye^{2x} = -1 + 2ce^{2x}$

(R) $x = 1 + xy\frac{dy}{dx} + \frac{\left(xy\frac{dy}{dx}\right)^2}{\lfloor 2} + ...$

$x = e^{xy\frac{dy}{dx}}$

$\log_e x = xy\frac{dy}{dx}$

$\int \frac{\log x}{x}dx = \int ydy \Rightarrow \frac{\log^2 x}{2} = \frac{y^2}{2} + c$

$y^2 = \log^2 x + c$

(S) $e^{x^2}ydy + e^{y^2}ydy + e^{x^2}x(y^2 - 1)dx = 0$

$e^{x^2}2ydy + (y^2 - 1)e^{x^2}2xdx + e^{y^2}2ydy = 0$

$d(e^{x^2}(y^2 - 1)) + d(e^{y^2}) = 0$

$e^{x^2}(y^2 - 1) + e^{y^2} = c$

16. (P) $2x^4ydy + y^2dy = x^2dx - 4x^3y^2dx$

$x^4(2y)dy + y^2(4x^3dx) + y^2dy - x^2dx = 0$

$d(x^4.y^2) + \frac{1}{3}d(y^3) - \frac{1}{3}d(x^3) = 0$

$x^4y^2 + \frac{y^3}{3} - \frac{x^3}{3} = c$

$3x^4y^2 + y^3 - x^3 = 3c$

(Q) Apply componendo & dividend

$\frac{e^x}{e^{-x}} = \frac{dx}{dy} \Rightarrow dy = e^{-2x}dx$

$y = \frac{-e^{-2x}}{2} + c$

$2ye^{2x} = -1 + 2ce^{2x}$

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(R) $x = 1 + xy\frac{dy}{dx} + \frac{\left(xy\frac{dy}{dx}\right)^2}{\lfloor 2} + ...$

$x = e^{xy\frac{dy}{dx}}$

$\log_e x = xy\frac{dy}{dx}$

$\int \frac{\log x}{x} dx = \int y dy \Rightarrow \frac{\log^2 x}{2} = \frac{y^2}{2} + c$

$y^2 = \log^2 x + c$

(S) $e^{x^2} y dy + e^{y^2} y dy + e^{x^2} x\left(y^2 - 1\right)dx = 0$

$e^{x^2} 2y dy + \left(y^2 - 1\right)e^{x^2} 2x dx + e^{y^2} 2y dy = 0$

$d\left(e^{x^2}\left(y^2 - 1\right)\right) + d\left(e^{y^2}\right) = 0$

$e^{x^2}\left(y^2 - 1\right) + e^{y^2} = c$

17. (P) $A = \int_0^{2\pi} y dx = \int_0^{2\pi} (1 - \cos\theta)(1 - \cos\theta) d\theta = 3\pi$

(Q) Let tangent $x\cos\alpha + y\sin\alpha = 1$

it's area is with coordinates axes

$A = \left|\frac{1}{2\cos\alpha\sin\alpha}\right| \Rightarrow A_{min} = 1$

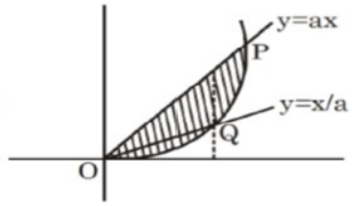
(R) Curve is rectangular Hyperbola

$(x + 2)(y - 1) = -3$

And pair of lines $(x + 2)(y - 1) = 0$ are pair of asymptotes.

$\therefore$ Area $= 2c^2 = 6$.

(S)



Points of intersection are

$P\left(4a^2, 4a^3\right), Q\left(4, \frac{4}{a}\right)$

Required area

$= \int_0^4 \left(ax - \frac{x}{a}\right)dx + \int_4^{4a^2} \left(ax - \frac{x^2}{4a}\right)dx$

$\alpha = \frac{16}{3a} - \frac{8}{a} + \frac{8a^5}{3}$ which is increasing

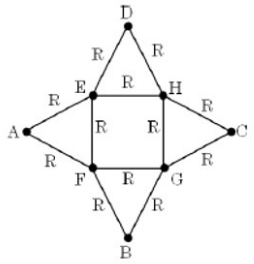
$\Rightarrow$ maximum value of area occurs of a = 2

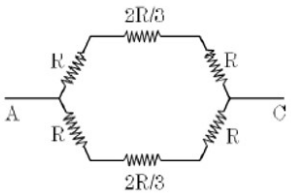
$\Rightarrow \alpha = 84$

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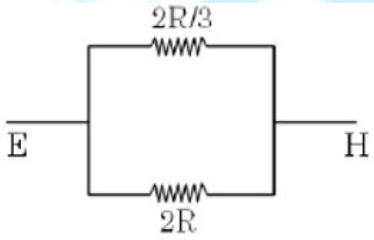
18. For R_{AC}





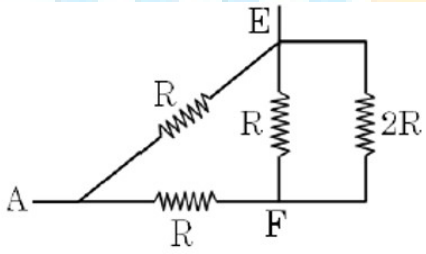
$$R_{AC} = \frac{4R}{3} = \frac{2}{3}\Omega$$

For R_{EH} :-



$$R_{EH} = \frac{R}{2} = \frac{1}{4}\Omega$$

For R_{AE} :



$$R_{AE} = \frac{5R}{8} = \frac{5}{16}\Omega$$

19. Since 10 divisions of Vernier scale coincide with 11 divisions of main scale, the Vernier scale reading can be considered from the rear.
For zero error MSR = -1, VSR = 3 (i.e. 7th mark)
Hence ZE = –0.7 mm
For measurement MSR = 11, VSR = 5
Hence measurement is 11.5 mm

20. Refraction at boundary brings object closer by $t\left[1-\frac{1}{\mu}\right]=10\left[1-\frac{1}{2}\right]=5$ cm

Hence object distance for lens is –15 cm

For focal length $\frac{1}{f}=\left[\frac{3}{2}-1\right]\left[\frac{2}{5}\right]$

$\Rightarrow f = 5\,cm$

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Hence $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}\Rightarrow\frac{1}{v}-\frac{1}{-15}=\frac{1}{5}$

$\Rightarrow\frac{1}{v}=\frac{3-1}{15}\Rightarrow v=\frac{15}{2}$

$m=\frac{v}{u}=\frac{\frac{15}{2}}{-15}=-\frac{1}{2}$

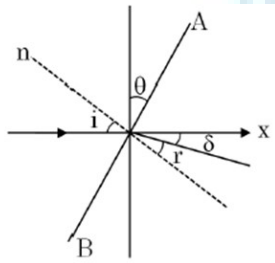
21. The plane surface is AB, n is normal to the surface.

Angle of incidence $i=\theta$

$\sin\theta=\mu\sin r$

$\Rightarrow\theta=\mu r$

$\therefore$ Deviation $\delta=\theta-r=\theta\left(1-\frac{1}{\mu}\right)$

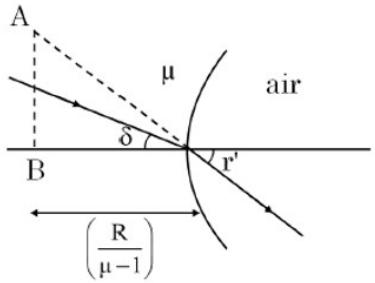


Now consider refraction at the curved surface

$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}\Rightarrow\frac{1}{v}-\frac{u}{-\infty}=\frac{1-u}{R}$

$\therefore v=\frac{R}{1-\mu}=-\left(\frac{R}{\mu-1}\right)$

$\therefore$ X co-ordinate of the image is $-\left(\frac{R}{\mu-1}\right)$



Snell's law gives $\mu\sin\delta=1.\sin'$

$\Rightarrow\mu.\delta=r'$

$\Rightarrow r'=(\mu-1)\theta$

Rays appear to be diverging from A.

$BA=|v|r'$

$=\left(\frac{R}{\mu-1}\right)(\mu-1)\theta=R\theta$

$\therefore$ Co-ordinates of point A are $\left(\frac{-R}{\mu-1},R\theta\right)$

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22. The electric field is given by

$$\vec{E} = \sqrt{5}E_0 \sin\left(\omega t + \frac{\omega z}{c}\right)\left(\frac{i+2j}{\sqrt{5}}\right)$$

Direction of propagation is

$$\hat{S} = -\hat{k} = \hat{E} \times \hat{B}$$

$$\Rightarrow \hat{B} = -\hat{k} \times \left(\frac{i+2j}{\sqrt{5}}\right)$$

$$= \frac{-\hat{j}+2\hat{i}}{\sqrt{5}}$$

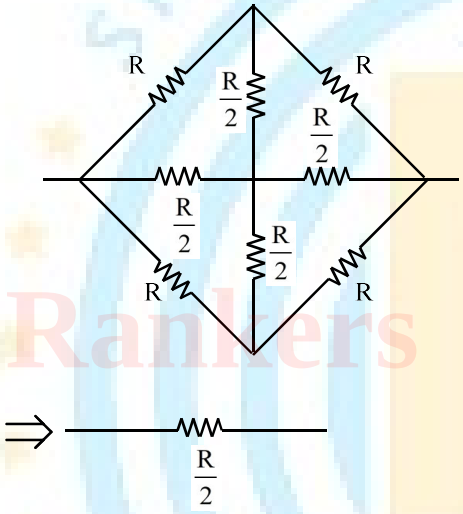
$$\Rightarrow \vec{B} = \frac{\sqrt{5}E_0}{C} \sin\left(\omega t + \frac{wz}{e}\right)\left(\frac{2i-j}{\sqrt{5}}\right)$$

23. Since the aperture is same for both lenses

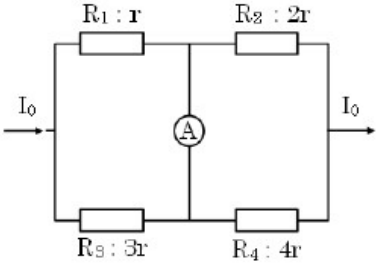
to make a telescope $f_0 > f_e$

to make a microscope $f_e > f_0$

24. Between any two opposite vertices the equivalent circuit according to folding method becomes



25.



Current through ammeter :-

$$\frac{I_0}{4r}.3r - \frac{I_0}{6r}.4r = \frac{3I_0}{4} - \frac{2I_0}{3} = \frac{I_0}{12}$$

$$\therefore I = \frac{1}{12}A$$

$$\therefore 24I = 2$$

26. For destructive interference

$$d\sin\theta = (2n+1)\frac{\lambda}{2}$$

$$\Rightarrow \sin\theta = (2n+1)\frac{6}{2\times 5}$$

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$=(2n+1)\frac{3}{5}$

$\Rightarrow \theta = \sin^{-1}\left(\frac{3}{5}\right) = 37°$

27. Due to thin glass slab image distance decreases by $t\left[1-\frac{1}{\mu}\right] = 0.9\left[1-\frac{25}{\quad}\right] = 0.3$ cm

$\therefore$ Final image distance is +11.7 cm

Applying les formula

$$\frac{1}{12}-\frac{1}{-240}=\frac{1}{11.7}-\frac{1}{-\mu_2}$$

$\Rightarrow \mu_2 = -492.63$

$\Rightarrow$ Distance moved = 252.63 cm

28. For maximum number of reflections the angle between mirrors must be smallest. Hence θ chosen would by 8°.

Using method of reflection of mirrors

No. of reflections $= \left[\frac{90}{8}\right] = 11$

29. The rays obey Snell's law is $\mu \sin\theta =$ constant

Where θ is angle made by ray at a given location with x-axis

By applying Calculus $\frac{1}{R} = \frac{\sin\theta}{n}\frac{dn}{dx}$

30. $\theta = \frac{1.22\lambda}{D}$

$D = 2R_0A^{1/3} = 2.44\times10^{-13}\times2$

$$\theta = \frac{1.22\times3\times10^{8}}{\frac{5}{4}\times10^{25}\times4\times1.22\times10^{-15}}$$

$$= \frac{3}{5}\times10^{-2}\text{ rad}$$

31. $P, S \rightarrow$ The electric field vectors at a given instant will be confined in an elliptical cross-section.

$Q \rightarrow$ The electric field vector at a given instant is confined in a circular cross-section

$R \rightarrow$ The $\vec{E}$ at a given instant will be confined in a linear cross-section.

32. P $\rightarrow$ Plane wave front reflects as a plane wave front near a plane mirror.

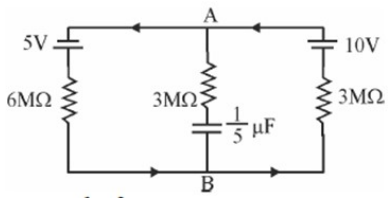
Q $\rightarrow$ Plane wave front becomes converging after passing through converging lens

R $\rightarrow$ Diverging wave front forms a virtual image (i.e. diverging wave front) after reflection from a convex mirror

S $\rightarrow$ Diverging lens forms virtual image for a real object, hence diverging wave front

33. Consider all possible combinations of phenomena that can happen as different rays propagate from the object.

34. For τ

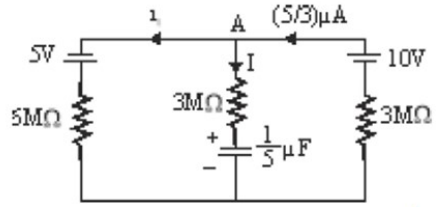


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$$R_{eq} = \frac{6\times 3}{6+3} + 3 = 5M\Omega$$

$$\tau = 5\times\frac{1}{5} = 1\sec$$

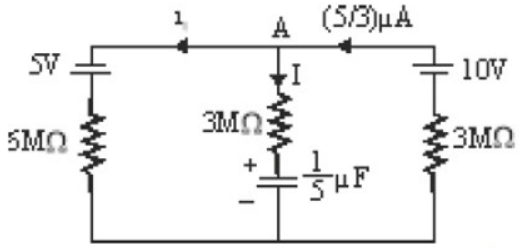
$\Rightarrow$ For q_{max} at steady state



$$R_{eq} = \frac{6\times 3}{6+3} + 3 = 5M\Omega$$

$$\tau = 5\times\frac{1}{5} = 1\sec$$

$\Rightarrow$ For q_{max} at steady state



Current passing through 10 V battery = $\frac{5}{3}\mu A$

$$V_A + 5 - \frac{5}{3}\times 6 = V_B$$

$$V_A - V_B = 5V$$

$$q_{max} = 1\mu C$$

$\Rightarrow$ at time t = 1 sec

$$q = 1\left(1 - e^{-1}\right) = 0.63\mu C$$

$$i = \frac{dq}{dt} = e^{-1} = 0.31\mu amp$$

From KVL

$$\frac{0.63}{1/5} + (0.37)3 + 5 - i_1(6) = 0$$

$$\mu = 1.54\mu A$$

CHEMISTRY

35. Conceptual

37. $CrCl_3.6H_2O$

$$\pi = iCRT \quad i = 1+\alpha(n-1)$$

$$98.4 = i\times 2\times 0.082\times 300$$

$$i = 2$$

$$1+\alpha(n-1) = 2$$

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$1+n-1=2(\alpha=1)$

$\boxed{n=2}$

$CoCl_3.4NH_3$ can be written as

$$\left[Co(NH_3)_4Cl_2\right]Cl \rightarrow \left[Co(NH_3)_4Cl_2\right]^{\oplus} + Cl^{\ominus}$$

$$Cl^{\ominus} + \underset{[AgNO_3]}{Ag^{\oplus}} \rightarrow AgCl\downarrow$$

$$\left[Co(NH_3)_4Cl\right]Cl + Ag^{\oplus} \rightarrow AgCl + \left[Co(NH_3)_4Cl_2\right]^{\oplus}$$

t = 0 $1\times2=2$ mole 1×2 0 0

$= 2$ mole

0 0 2 mole 2 mole

Weight of AgCl = $2\times143.5=287$ g

Final Solution = 2 mole $\left[Co(NH_3)_4Cl_2\right]NO_3$

Total volume = 1 + 2 = 3 litre

$$C=\frac{2}{3}$$

$\pi = iCRT$

$$=2\times\frac{2}{3}\times0.082\times300$$

$\pi = 32.8$ atm

$i=1+\alpha(n-1)$

$=1+\alpha(2-1)$

$= 1 + \alpha$

$= 1 + 1$

$i = 2$

40. pH = 5 pH = 6

$[H^+]=10^{-5}$ $[H^+]=10^{-6}$

On decreasing $[H^+]$ concentration by 10 times, the half-life of reaction increases by 10 times that means reaction rate has been decreased by 10 times. Therefore, we can say that the order of reaction with respect to $[H^+]$ concentration is one.

41. A B + C

100 – x x x

100 + x = 140

x = 40

A = 60 mm

$$K=\frac{2.303}{30}\log_{10}\frac{100}{60}$$

$$K=\frac{2.303}{30}\times(1-0.8)$$

$$K=\frac{2.303\times0.2}{30}$$

$$T_{1/2}=\frac{2.303\times0.3}{2.303\times0.2}\times30=45\,\text{min}$$

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44. $\frac{P_O - P_S}{P_O} = X_{solute}$

$\therefore \frac{100-98.7}{100} = X_{solute} = 0.013$

Now, $\Delta T_f = K_f \times m$

$0.72 = K_f \times \frac{0.013 \times 1000}{0.978 \times 78}$

$\therefore K_f \approx 4.2\,K\,Kg/mol$

$\therefore$ Reported answer = 4

45. $n_{eq} I_2 = n_{eq} SO_2 + n_{eq}$ hypo

$25 \times 1 \times 10^{-3} = n \times 2 + 10 \times 1 \times 10^{-3} \times 1$

$\therefore n_{SO_2} = 7.5 \times 10^{-3}$

Hence, $4\,hr = 2 \times t_{1/2}$

47. (i), (ii), (iii), (v), (vi), (ix)

50. No. of particles $\downarrow$, so vapour pressure, i$\downarrow$

$DT_f = mK_f i$

$DT_f \downarrow$, freezing point $\uparrow$

51. (P) $K = \frac{K_A}{2} = \frac{0.693}{80} s^{-1}$

$r = K[A]_0 = \frac{0.693}{80} \times 0.8$

$= 6.93 \times 10^{-3} M/s$

$(Q) r_A = \frac{(4-2)}{40 \times 5} = 10^{-2} M/s$

$r = \frac{r_A}{2} = 5 \times 10^{-3} M/s$

$(R) r = K[A]$

$= \frac{0.693}{80} \times 0.04$

$= 3.465 \times 10^{-4} M/s$

$(S) = KA = \frac{1}{t_{1/2}} \times \frac{1}{[A]_0} = \frac{1}{40} \times \frac{1}{4/5}$

$r = \frac{5}{40 \times 4 \times 2} \times (0.008)^2 = 10^{-6} M/s$

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